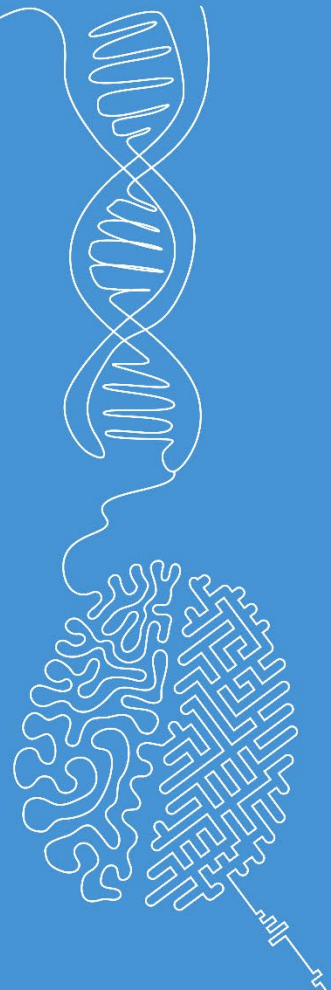


X-Ray

Imaging for the Life Sciences

Robert Vorburger



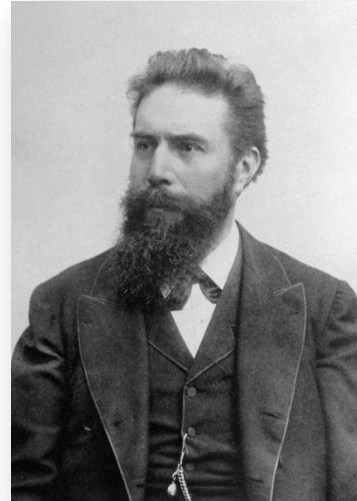
Content

- History
- Principles
- Applications
- Data Aspects



Content

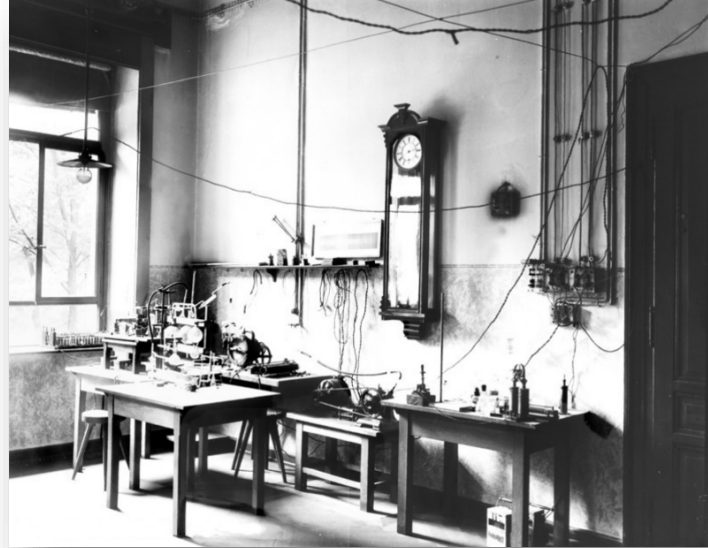
- **History**
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Wilhelm Conrad Röntgen



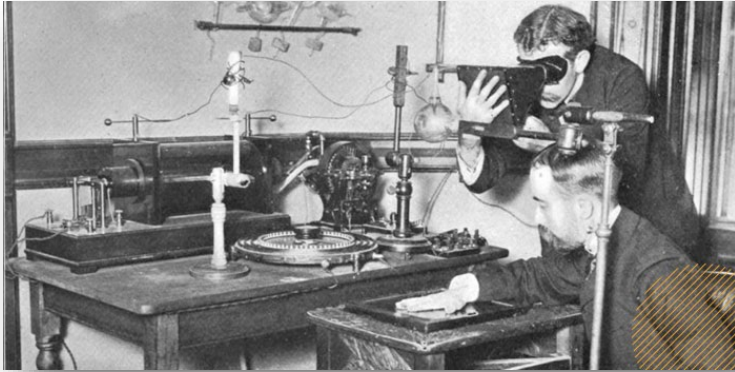
Wilhelm Conrad Röntgen
(1900)



Röntgen's laboratory in the former Physical
Institute of the University of Würzburg
(1895)

← year of discovery

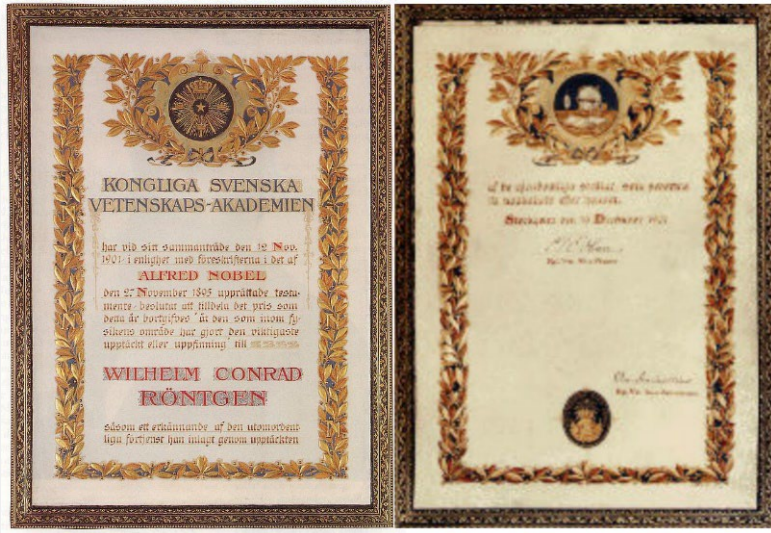
Wilhelm Conrad Röntgen



Albert von Kölliker's hand (1896)

He was a Swiss medical scientist from Zurich pioneering histology, who also taught at University of Zurich.

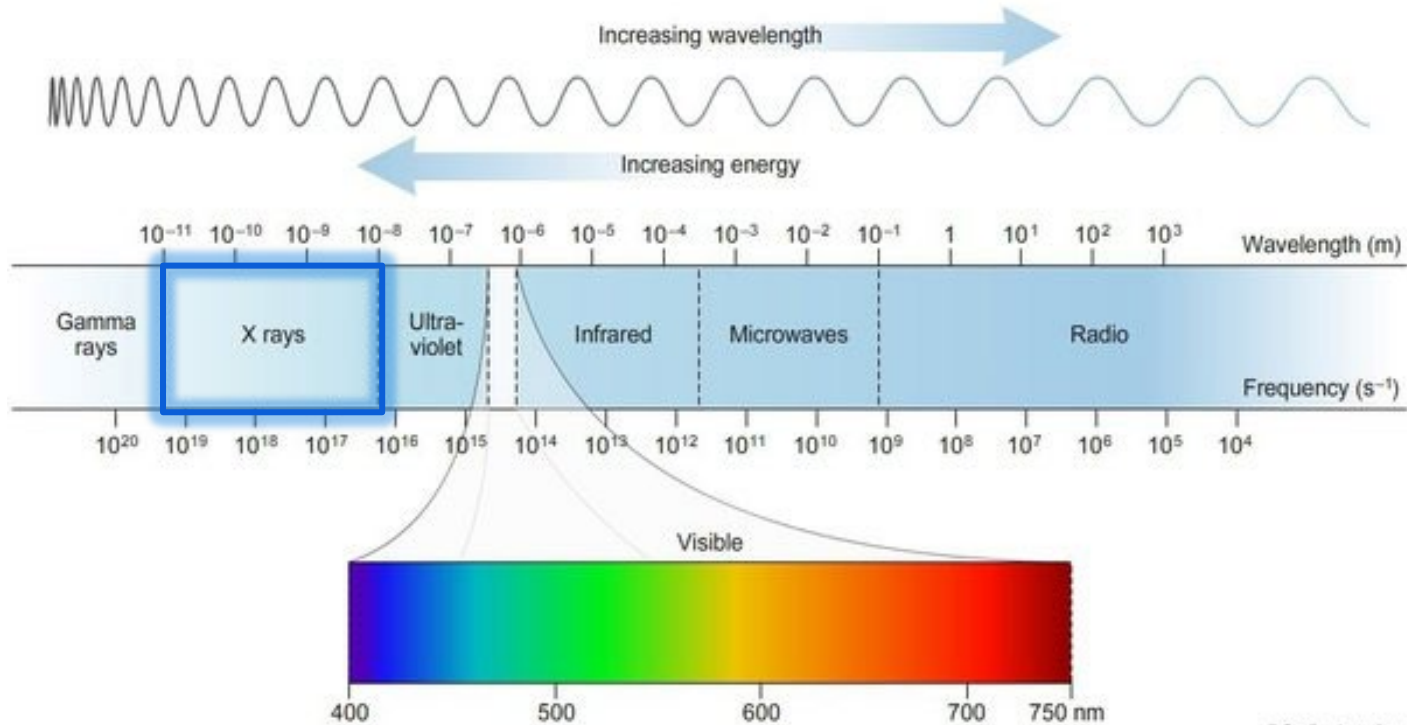
Wilhelm Conrad Röntgen



Nobel prize in physics 1901

First ever awarded Nobel prize. Röntgen did not patent his invention on purpose to make it available to everybody and he did not accept an offered title of nobility.

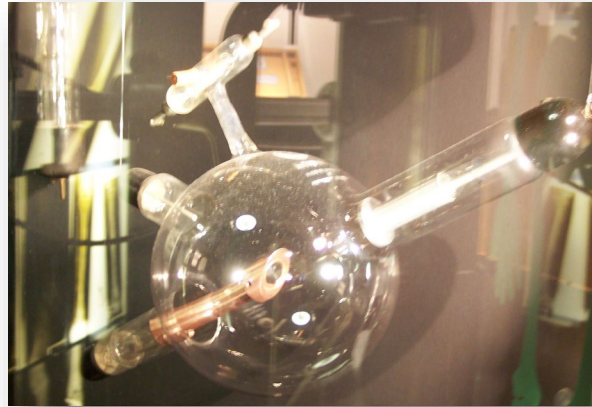
What are X-Rays?



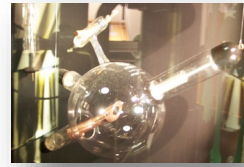
© Sapling Learning

Content

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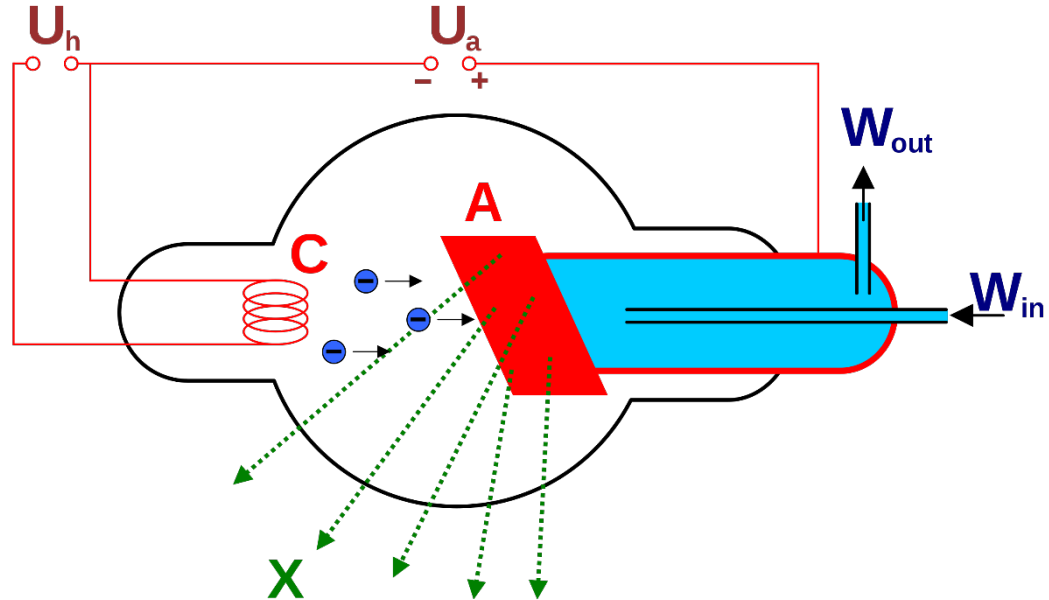


X-Ray Generation



X-rays are generated using a X-ray tube

1. Cathode (C): heating of coil (heat voltage U_h) leads to emission of electrons (-)
2. Electrons can freely travel through vacuum in glass capsule (black line)
3. Anode (A): electric field between C and A (anode voltage U_a) attracts electrons
4. Impact of electrons onto A lead to emission of X-rays (X)
5. Water cooling (W_{out}/W_{in})

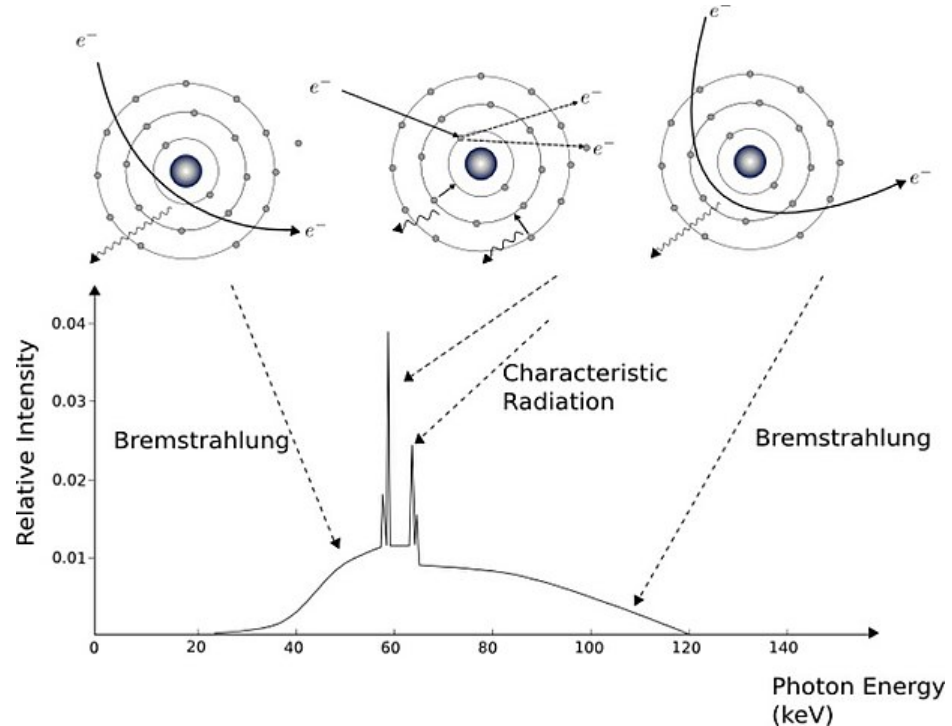


X-Ray Generation

The emitted X-rays consist of a range of different energies

Two types of radiation:

- **Bremsstrahlung** (continuous line)
flying electrons are slowed down by attractive force of positive nucleus
- **Characteristic radiation** (peaks)
Flying electrons hit electrons in electron shell of anode
 - creation of «hole»
 - filling of «hole»
 - emitting of X-rays

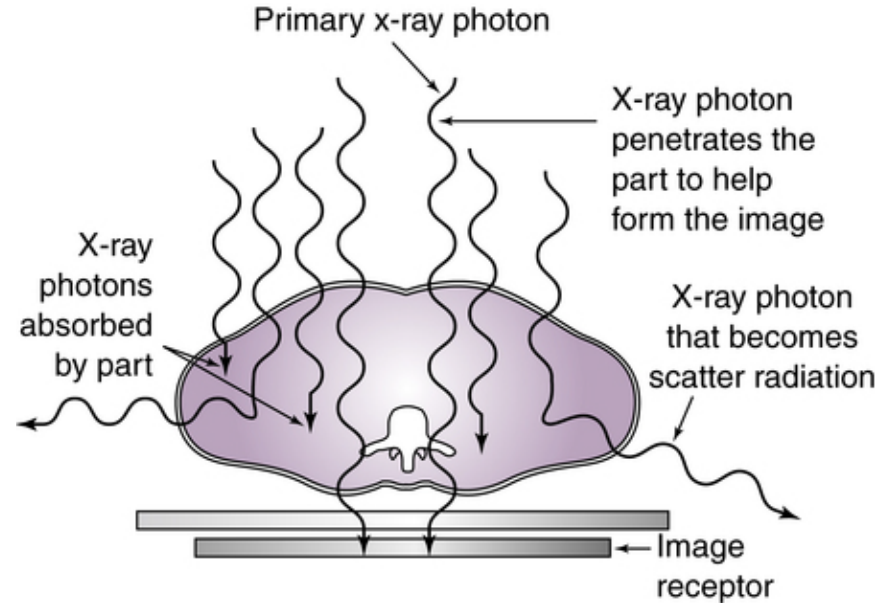


X-Ray Absorption

Emitted X-rays interact with target/part

- Transmission: photons passing through the body
- Attenuation:
 - **Absorption**: partial or total absorption of photon
 - **Scatter**: photons diverted in a new direction

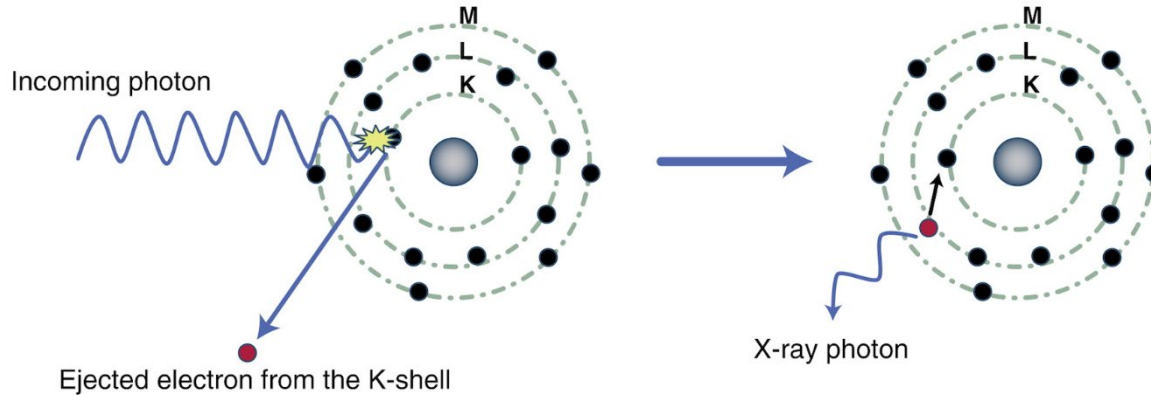
Note: a photon is a (quasi-)particle with a defined energy and impulse consisting of light (which is also a wave...)



X-Ray Absorption

Photoelectric effect

- X-ray photons hit electrons in electron shell of target: creation of «hole»
- Filling of «hole» by another electron
- Emission of light (characteristic X-ray)
- Characteristic X-ray is (typically) absorbed in target



Full attenuation, but relatively rare

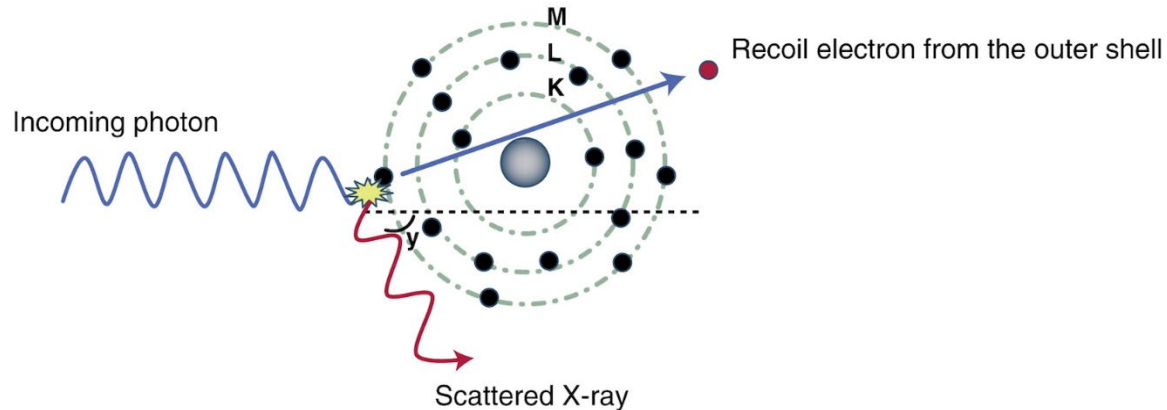
Note: Albert Einstein received Nobel prize in physics for the discovery of the photo effect in 1921

X-Ray Absorption

Compton effect

- Incident X-ray photon hits electron in the shell of target atom and transfers some energy
- Electron gets ejected from electron shell
- X-ray photon is diverted in (a new) direction with reduced energy

Partial attenuation, relatively frequent



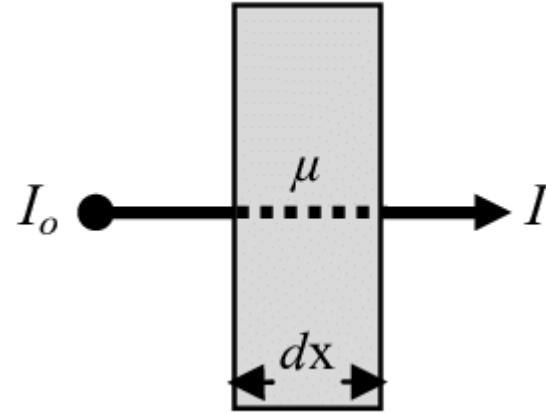
X-Ray Absorption

Beer-Lambert's law

Various absorption processes can be phenomenologically described using a simple law

- Linear attenuation coefficient (of material) μ , which depends on X-ray photon energy E : $\mu(E)$
(Combines Photoelectric and Compton effect)
- Target thickness: x
- Intensity: I

$$\mu(E) = -\frac{\ln\left(\frac{I}{I_0}\right)}{dx} = \frac{1}{dx} \ln\left(\frac{I_0}{I}\right)$$



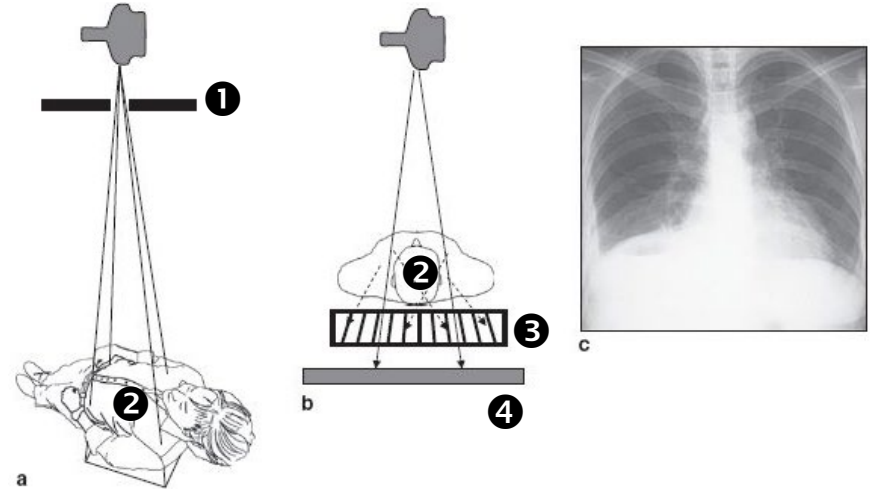
$$I = I_0 e^{-\mu(E)dx}$$

Contrast is generated by different absorption strengths of different constituents (mainly bone)

Projection Imaging

Set-up

1. Collimator
narrows beam to prevent unnecessary radiation of parts outside target
2. Target
X-rays which pass through the target can be detected
3. Anti-scatter grid
suppresses (Compton) scattered X-rays
4. Detector
solid-state detectors (historically: film)



Result: two-dimensional projection of the target as negative (attenuation image)
(where $\mu(E) = 0$ is black and the highest $\mu(E)$ value is white)

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Clinical Applications

- Extremity X-Ray



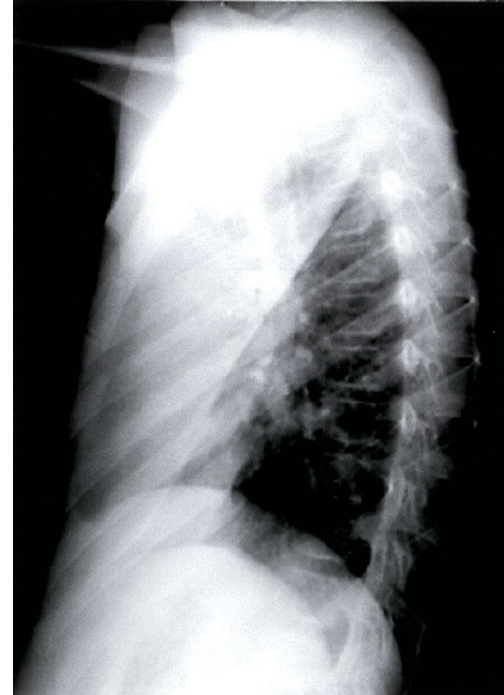
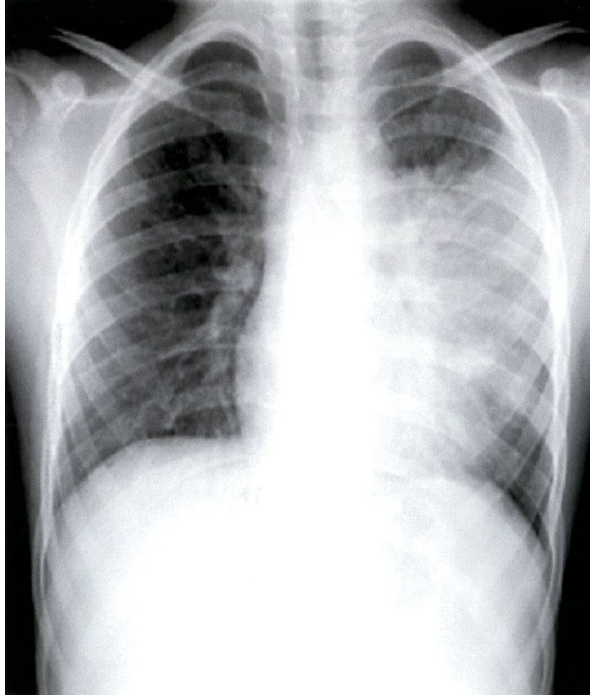
Clinical Applications

- Extremity X-Ray



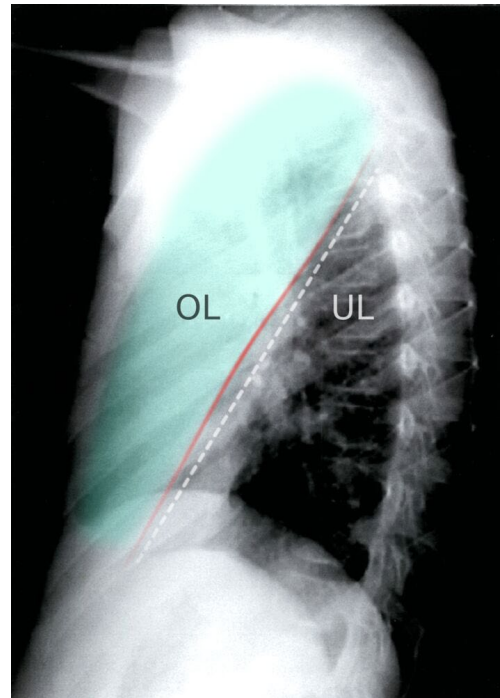
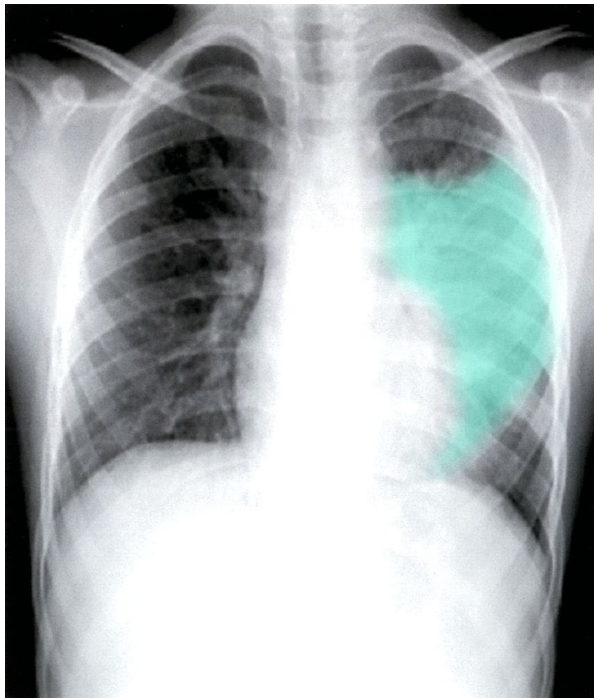
Clinical Applications

- Thorax X-ray



Clinical Applications

- Thorax X-ray



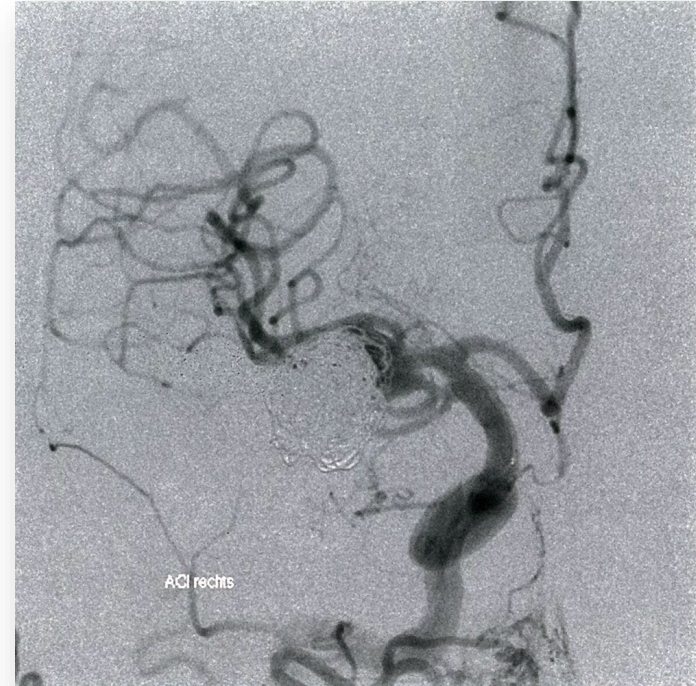
Clinical Application

Digital subtraction angiography

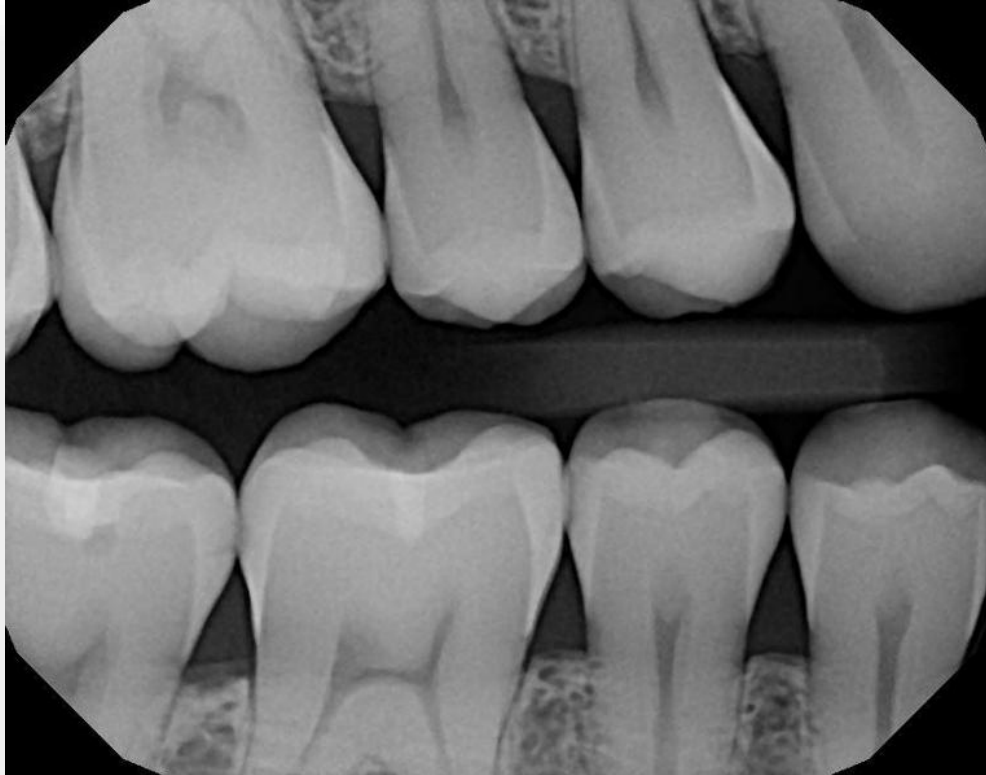
Contrast agent-based technique to image blood vessels (angiography)

1. Acquisition of a standard (native) X-ray image
2. Injection of contrast agent (Iodine-based)
3. Acquisition of contrast-enhanced X-ray image
4. Subtraction of both images

Note: contrast agents (Barium-based) are also used without subtraction

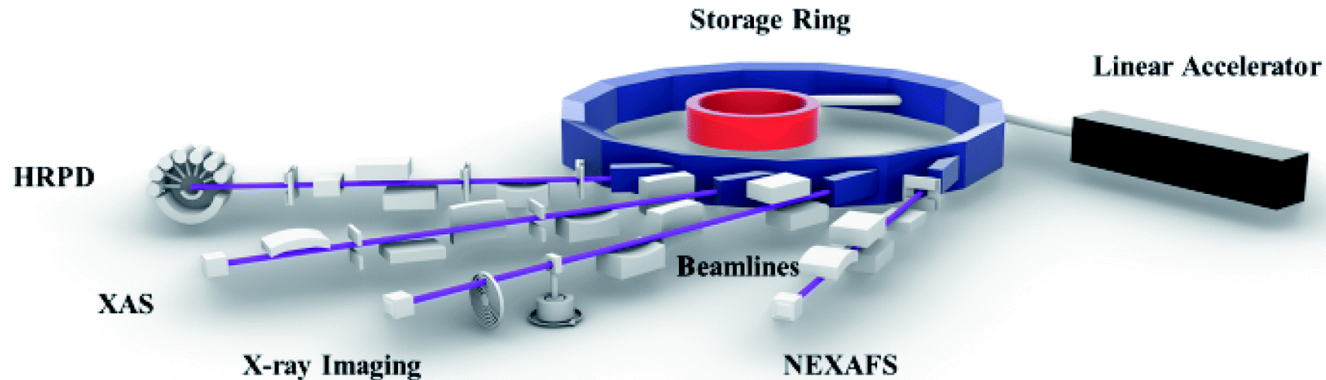
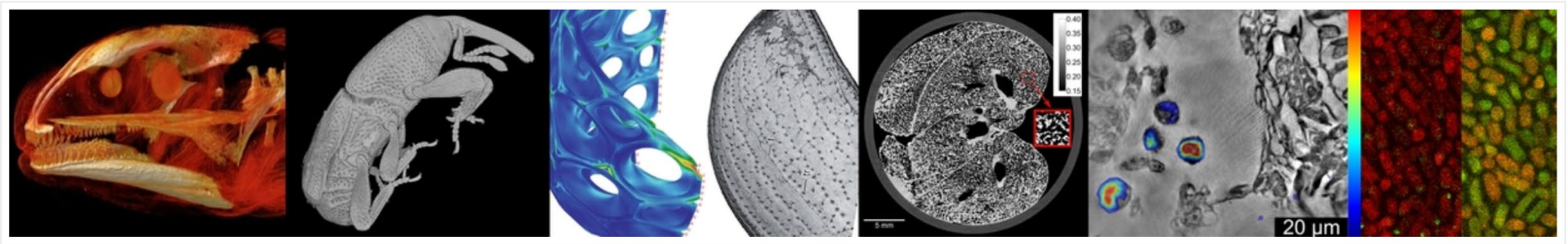


Dental Application



Synchrotron X-Ray

X-ray Imaging of Small Organisms, Tissues, and Cells with Synchrotron Radiation at KIT: Overview and Selected Examples



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Image Properties

- **Usage:** frequently used for numerous pathologies, because of availability, economy and despite (relatively little) radiation
- **Projections:** typically two images are acquired, one from the side (sagittal) and one from the front/back, also there are special angulations for specific pathologies
- **Distortions:** apparently different organ sizes depending on imaging geometry
 - Patient can stand: X-rays passing from back to front (posterior-anterior)
 - Patient lays in bed (e.g. intensive care): X-rays passing from front to back (anterior-posterior)
- **Contrast:**
 - High contrast for bone and calcified structures
 - Low soft tissue contrast
- **Noise:** relatively low (Poisson-distributed) noise
- **Spatial resolution:** (slight) blurring of target on image, because of
 - Distance of target from detector
 - Detector properties

